

Closed-Loop Streaming Synthesis Bounded-Drift, Arbitrary-Length Audio–Video Generation by Treating Chunk Hand-off as Feedback Control

Aleksander Majda

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Abstract

Generating arbitrarily long videos with diffusion transformers requires splitting the temporal dimension into overlapping chunks. Naive open-loop chunk-streaming accumulates exposure-bias drift: each chunk is conditioned on the previously generated latent, compounding distributional errors and causing scene collapse or grain amplification after a few hundred frames. We present **Closed-Loop Streaming Synthesis (CLSS)**, a suite of four post-denoising corrections that close the feedback loop between successive chunks without modifying the transformer weights: calibrated context re-noising, per-channel EMA-tracked AdaIN, frequency-band soft shrinkage, and a dynamic anchor bank. CLSS is implemented as a ComfyUI custom-node pipeline targeting LTX-2.3 22B (a 44 GB audio-video diffusion transformer) on consumer 16 GB VRAM hardware via GGUF 4-bit quantisation and CPU block-streaming. Generation is driven by a reference-parity multi-modal guider (split per-modality classifier-free guidance, modality-isolation guidance, and spatiotemporal guidance via a value-passthrough self-attention skip) that correctly handles ComfyUI’s packed audio-video latent representation, and all streaming noise is seeded for bit-reproducible runs. A two-stage pipeline generates a draft audio-video via CLSS streaming and then refines it using a 3-step distilled-LoRA schedule at $2\times$ spatial resolution.

We report matched text-to-video (T2V) and image-to-video (I2V) experiments: single-scene, ≈ 53 -second clips generated as ten chunks ($\text{new_lf} = 16$, $\text{overlap} = 8$) under an identical seed and guidance stack (split per-modality CFG 4.0 video / 7.0 audio, STG 1.0 on both modalities at block 28, rescale 0.70; three transformer passes per step), differing only in the presence of a guide image. Stage 1 holds video boundary cosine similarity at 0.92–0.97 in I2V; a two-band spatial detail anchor holds the high-frequency energy share flat across the full run in both modes ($0.510 \rightarrow 0.510$ I2V, $0.512 \rightarrow 0.520$ T2V),

closing the progressive-softening channel, and a reference-exact RMS anchor holds audio loudness to its chunk-1 level throughout ($\pm 5\%$) against raw per-chunk drifts of up to $+72\%$ (T2V). Stage 2 ($2\times$ resolution, 3-step distilled refinement in four windows) refines video against frozen Stage-1 audio, with its own per-window detail anchor referenced to the upscaled Stage-1 slice; end-of-window fidelity to Stage-1 structure stays at 0.93–0.97 (I2V). Two failure modes are characterised rather than solved: a loudness-envelope repetition intrinsic to chunked autoregressive audio over a static prompt (shown immune to continuity-buffer, guidance-weight, and reference-perturbation interventions), and audio high-frequency energy growth over the run. Total generation time is ≈ 98 minutes per clip in either mode on a single 16 GB GPU.

1 Introduction

Modern video diffusion transformers operate on patchified latent representations where temporal resolution is down-scaled $8\times$ relative to pixel frames. Fitting a 30-second clip (81,840 tokens at 22×40 spatial) into a single forward pass requires very large GPUs. The standard remedy is *temporal chunking* via a Streaming Latent Buffer (SLB): generate the video in short overlapping segments, conditioning each on the previous denoised output. Section 2 reviews this mechanism and the exposure-bias problem it introduces.

CLSS closes the feedback loop between chunks by applying five lightweight post-denoising corrections that together bound the closed-loop gain below one, preventing drift from accumulating without modifying the transformer weights. The corrections are applied purely at the boundary between chunks and add negligible computational overhead.

Contributions.

1. Five closed-loop correction modules for temporal chunk streaming: calibrated context re-noising, EMA-tracked per-channel AdaIN, frequency-band soft shrinkage, a dynamic anchor bank, and a two-band spatial detail anchor that closes the progressive-softening channel the one-sided shrinkage leaves open (Sections 3.1–3.5).
2. A linear stability analysis of the feedback loop in latent space, including an empirical per-chunk gain measurement protocol (Section 3.6).
3. A memory-efficient two-stage pipeline for LTX-2.3 22B on 16 GB VRAM using GGUF 4-bit quantisation and CPU block-streaming (Section 5).
4. A reference-parity multi-modal guider for ComfyUI (split per-modality CFG, modality-isolation guidance, and STG via value-passthrough self-attention skip) that correctly handles ComfyUI’s packed audio-video latent representation (Section 5.5).
5. ComfyUI custom nodes implementing the full pipeline with seeded, bit-reproducible streaming noise and per-chunk reproducibility telemetry: unconditional settings dumps, per-frame origin/layout drift event tables, and a phase-lock repetition detector (Sections 5.4, 7.6).

6. Matched ≈ 53 -second T2V and I2V generations, identical seed and guidance stack, streamed as ten chunks each, isolating the effect of visual conditioning on continuity and repetition, and characterising the residual failure modes at this horizon — including a loudness-envelope repetition shown by intervention to be structural to chunked autoregressive audio rather than tunable by any conditioning or guidance knob (Sections 6, 7.6).

2 Background

2.1 LTX-2.3 Architecture

LTX-2.3 is a 22-billion-parameter diffusion transformer for joint audio-video synthesis. It operates on a latent space with an $8\times$ temporal downscale (video VAE, causal) and a $160\times 4\times$ audio downscale (audio VAE, 16 kHz, hop length 160, compression factor 4), yielding a video latent rate of $25/8 = 3.125$ lf/s and an audio latent rate of $16000/(160\times 4) = 25.0$ lf/s. The model accepts interleaved video and audio tokens together with Gemma-3 text embeddings and is trained with flow-matching on the rectified-flow schedule.

2.2 Streaming Latent Buffer (SLB)

The SLB provides temporal continuity in chunk streaming. When generating chunk N , the last `overlap_lf` latent frames from chunk $N - 1$ are prepended to the new chunk’s latent, and the denoiser is conditioned on them at noise level τ_c :

$$\tilde{L}_{\text{overlap}} = (1 - \tau_c) \hat{L}_{\text{overlap}} + \tau_c \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, I) \quad (1)$$

Setting $\tau_c = 0$ yields maximal continuity but maximal drift risk; $\tau_c = 1$ decouples the chunks entirely. In LTX’s framework this is implemented as `VideoConditionByLatentIndex` with `strength = 1 - \tau_c`.

Implementation note: masked timestep, not a one-shot draw. Eq. (1) describes the target distribution of the overlap context, but the sampler does not literally draw ε once and hand the model a truncated schedule from τ_c . Tracing the actual mechanism (`ComfyBaseModel.process_timestep` for the LTX model, which the `noise_mask/strength` produced by `VideoConditionByLatentIndex` feeds into) shows that the per-token timestep supplied to the transformer at every sampling step t is $\tau_{\text{eff}} = (1 - \text{strength}) \cdot \sigma_t = \tau_c \sigma_t$ for overlap tokens, versus σ_t for the rest of the sequence — a *persistent* per-step rescaling of the global noise schedule, applied at every step, not a single noise-injection event applied once before an otherwise-standard remaining schedule. For a *linear* (rectified-flow) interpolation path, $x_t = (1 - t)x_0 + t\varepsilon$, these two descriptions coincide exactly: scaling every step’s effective time by a constant factor τ_c is algebraically identical to drawing x_{τ_c} once via Eq. (1) and continuing the *same* linear ODE from τ_c down to 0. LTX-2.3 is trained with a rectified-flow objective (Section 2), so the equivalence holds for this model; it would not hold in general

for a non-linear (e.g. cosine or EDM) noise schedule, where per-step time-rescaling and one-shot truncated-schedule sampling are not interchangeable.

2.3 Exposure Bias in Streaming Diffusion

Let $\hat{L}_N$ be the denoised latent of chunk N and $f_\theta(\cdot \mid \hat{L}_{N-1})$ the denoising operator conditioned on the previous chunk. In open-loop streaming:

$$\hat{L}_N = f_\theta(\hat{L}_{N-1} + \varepsilon_N) \quad (2)$$

The transformer was trained on latents drawn from the data distribution, but at inference it receives denoised latents from prior chunks, which lie off-manifold relative to its training distribution. Any distributional shift $\delta_N = \hat{L}_N - L_N^*$ (where L_N^* is the “on-manifold” latent for the same content) is in general amplified by the open-loop gain of the denoising process. Formally, if $g = \|\partial f_\theta / \partial L\|$ evaluated at L_{N-1}^* , then $\|\delta_{N+1}\| \approx g \|\delta_N\|$. When $g > 1$ this gives $\|\delta_N\| \sim g^N \rightarrow \infty$.

In practice g is not a global constant: it depends on the noise level schedule, step count, guidance strength, and the spectral content of the latent. Section 3.6 describes how g is estimated empirically per chunk via a perturbation measurement.

3 Closed-Loop Streaming Synthesis

CLSS adds five closed-loop correction modules that together bound the error accumulation by ensuring $\rho_{\text{closed}} = g \cdot (1 - \beta) \cdot (1 - \tau_c) < 1$.

3.1 Calibrated Context Re-Noising

The overlap latent fed as context to chunk N is re-noised to level τ_c via Eq. (1). This forces the denoiser to actively re-project the boundary context onto the data manifold rather than accepting it verbatim, partially correcting accumulated drift in the SLB.

In practice $\tau_c = 0.05$ provides a strong temporal constraint (boundary cosine similarity > 0.97) while allowing enough distributional repair to prevent drift accumulation. Larger values ($\tau_c = 0.20$, as suggested by some prior work) loosen the constraint too much for 22B GGUF streaming, causing visible motion discontinuities.

3.2 EMA-Tracked Per-Channel AdaIN

A slow exponential moving average (EMA) with rate $\lambda \approx 0.10$ tracks per-channel mean μ_c and standard deviation σ_c across chunks:

$$\mu_c^{(N)} = (1 - \lambda) \mu_c^{(N-1)} + \lambda \hat{\mu}_c^{(N)} \quad (3)$$

$$\sigma_c^{(N)} = (1 - \lambda) \sigma_c^{(N-1)} + \lambda \hat{\sigma}_c^{(N)} \quad (4)$$

where $\hat{\mu}_c^{(N)}, \hat{\sigma}_c^{(N)}$ are the per-channel statistics of chunk N ’s new frames. After generation, the overlap region is dropped and *only the new frames* L_N^{new} (Algorithm 1,

line 5) are blended toward the EMA reference via AdaIN with blend factor β :

$$\tilde{L}_N^{\text{new}} = (1 - \beta) L_N^{\text{new}} + \beta \text{AdaIN}(L_N^{\text{new}} \rightarrow \mu^{(N-1)}, \sigma^{(N-1)}) \quad (5)$$

The overlap region itself is never re-touched by AdaIN — it was already corrected as *previous* chunk’s L_{N-1}^{new} before being pushed to the SLB. This suppresses fast per-chunk statistical drift while allowing slow intentional scene evolution (lighting changes, colour grading) to pass through.

Two safety caps prevent over-correction:

- **Sigma max drift:** the EMA σ_c is clamped to at most $(1 + \Delta) \times \sigma_c^{(0)}$ (default $\Delta = 0.05$), preventing AdaIN from amplifying late-chunk residual noise.
- **AdaIN max amplification:** per-channel upward scaling is capped at $1.2\times$ the current chunk’s std (default), preventing AdaIN from boosting noise in channels that have become quiet.

3.3 Frequency-Band Soft Shrinkage

The latent is decomposed into $B = 4$ frequency bands via 3-D FFT along the (frame, height, width) axes. Each band b receives a continuous gain:

$$g_b^{(c)} = \min\left(1, \left(\frac{E_{\text{ref},b}^{(c)}}{E_b^{(c)}}\right)^{\gamma_b}\right) \in (0, 1] \quad (6)$$

where $E_b^{(c)}$ is the mean squared amplitude of band b in channel c , $E_{\text{ref},b}^{(c)}$ is the corresponding EMA reference, and γ_b is the per-band shrinkage exponent.

Only bands with excess energy relative to their EMA are attenuated. Bands with $\gamma_b = 0$ (the DC band) are never attenuated. Default exponents: $\gamma = (0.0, 0.05, 0.2, 0.3)$, which are conservative enough to preserve scene structure while suppressing temporal flicker.

The EMA is updated with the *corrected* (post-gain) energies, not the raw ones, so the reference tracks intentional scene content rather than accumulated error.

3.4 Dynamic Anchor Bank

A small bank of $M \leq 8$ keyframe anchors provides long-range visual identity conditioning. The bank is seeded from the *last* frame of chunk 1 (not the first) and updated when two consecutive chunks trigger a scene-change signal (cosine similarity below threshold $\rho_a = 0.78$ for $P = 2$ consecutive chunks). Additionally, a *forced insertion* every F_a chunks (chunk_index mod $F_a = 0$, chunk 1 always seeds regardless) prevents bank stagnation on visually stable sequences where the similarity threshold is never triggered. The reference default is $F_a = 5$; the ComfyUI node auto-derives $F_a = \max(2, \min(5, \lceil \text{num_chunks}/4 \rceil))$, giving $F_a = 3$ for the ten-chunk experiments in Section 6.

Anchors are retrieved via cosine similarity and injected as `VideoConditionByKeyframeIndex` at frame index 0 of the current chunk. Anchors too similar to the current overlap

(sim > 0.85) are skipped to avoid duplicating the SLB conditioning. LRU eviction discards the least recently used anchor when the bank is full.

3.5 Two-Band Spatial Detail Anchoring

The shrinkage of Section 3.3 is deliberately one-sided (Eq. (6) only ever attenuates), and this asymmetry has a measured long-run consequence: low-band energy inflates while high bands are only ever shrunk, so the *high-frequency share* of latent energy falls monotonically — +19.6% band-0 inflation over seven chunks in one measured run, with the uncapped band EMA itself drifting upward and legitimising the excess. Nothing in the shrinkage loop can *restore* band energy, and the perceptual result is progressive detail loss.

The detail anchor closes exactly this channel. Each frame is split into a spatial low band (a 3×3 average pool over the latent grid) and its high-band residual, and the mean squared energy of each band is equalised toward a *fixed* scene-first reference (captured at chunk 1, re-baselined on scene change) with symmetric per-chunk gains:

$$g_{\text{lo}} = \left(\frac{E_{\text{lo}}^{\text{ref}}}{E_{\text{lo}}} \right)^{1/2} \text{ clamped to } [0.90, 1.10], \quad g_{\text{hi}} = \left(\frac{E_{\text{hi}}^{\text{ref}}}{E_{\text{hi}}} \right)^{1/2} \text{ clamped to } [0.90, 1.12] \quad (7)$$

Unlike the shrinkage EMA, the reference here does not track the output — drift cannot legitimise itself — and unlike the shrinkage gains, these are symmetric: lost high-band energy is restored, not merely excess removed. The per-chunk clamp bounds the correction so a genuine scene change cannot be fought harder than ± 10 – 12% per chunk, and a $\pm 0.5\%$ dead-band makes the module a strict no-op when energies already match. The same mechanism runs per-window in Stage 2, referenced there to the upscaled Stage-1 slice of the same window (Section 5.2). In the reported runs it holds the Stage-1 high-frequency share flat end-to-end ($0.510 \rightarrow 0.510$ I2V, $0.512 \rightarrow 0.520$ T2V; Section 6).

3.6 Stability Analysis

The AdaIN blend (Section 3.2) and context re-noising (Section 3.1) together reduce the magnitude of any per-chunk latent perturbation by a factor $(1 - \beta)(1 - \tau_c)$ before it reaches the next chunk’s denoiser. This gives the *loop gain*:

$$\rho_{\text{loop}} = (1 - \beta) \cdot (1 - \tau_c) \quad (8)$$

The combined closed-loop gain, accounting for the model’s open-loop amplification g , is:

$$\rho_{\text{closed}} = g \cdot \rho_{\text{loop}} \quad (9)$$

Bounded drift requires $\rho_{\text{closed}} < 1$. With $\beta = 0.40$ and $\tau_c = 0.05$, $\rho_{\text{loop}} = 0.57$, providing $\approx 43\%$ gain reduction per chunk before accounting for g .

Why the two factors multiply. $(1-\tau_c)$ and $(1-\beta)$ act on disjoint tensor regions at different points in the cycle — $(1-\tau_c)$ on the *overlap* region when it is *read* as context (Section 3.1), $(1-\beta)$ on the *new-frame* region when it is *written* as output (Section 3.2, and see the corrected notation there: AdaIN never re-touches the overlap). This is not two attenuators applied to the same signal at the same place; it is two attenuators the same error signal passes through exactly once per loop iteration: a perturbation in chunk N ’s output is written with gain $(1-\beta)$ (since AdaIN is applied uniformly across L_N^{new} , including the tail frames that become the next SLB), then read back with gain $(1-\tau_c)$ once it becomes chunk $N+1$ ’s overlap context. For a linear, time-invariant loop this is exactly the standard construction of a loop gain from attenuators in series — the product is valid *because* each attenuator sits on the one path the error must traverse per cycle, not because they act on the same coefficients. What Eq. (8) does not capture is any cross-term where the correction strength itself depends on chunk content (e.g. the AdaIN amplification cap, Section 3.2, engages non-uniformly); that content-dependence is exactly why g cannot be folded into a single constant either (below).

Note that Eq. (8) is a first-order approximation that treats the two corrections as independent multiplicative attenuators. In practice g is not constant: it depends on the step count, guidance strength, noise schedule, and the spectral properties of the current latent. The analysis therefore provides a stability condition rather than a precise quantitative bound. The frequency-band shrinkage (Section 3.3) provides additional spectral attenuation beyond ρ_{loop} , especially in high-frequency bands, but is not included in the linear gain model because its contribution is content- and chunk-dependent.

Hyperparameter Choices. $\beta = 0.40$ is the largest value that preserves scene-change responsiveness in practice: at $\beta \geq 0.5$ slow colour shifts accumulate due to excessive EMA anchoring. $\tau_c = 0.05$ is a conservative re-noising level that maintains high boundary similarity (> 0.97) while partially correcting SLB distribution shift; larger values ($\tau_c \geq 0.20$) cause visible motion discontinuities on 22B streaming. Together they yield $\rho_{\text{loop}} = 0.57$, which empirically provides adequate margin given the band-energy-growth proxy for g discussed in Section 7 (note: not the g_{SLB} perturbation measurement below, which was not exercised in the reported runs). $\lambda = 0.10$ (EMA rate) was chosen to respond to gradual scene evolution without over-weighting individual chunks; $\gamma = (0.0, 0.05, 0.2, 0.3)$ are conservative exponents that attenuate only bands with clear excess energy.

Per-Chunk Gain Measurement. g is measured empirically by running a second denoising pass with a slightly perturbed overlap latent $\tilde{L}_{\text{overlap}} + \delta$ ($\|\delta\|_F = \epsilon \|\tilde{L}_{\text{overlap}}\|_F$, $\epsilon = 0.01$) and comparing the output perturbation to the input:

$$g_{\text{SLB}} = \frac{\|\Delta_{\text{output}}^{\text{SLB}}\|_F}{\|\delta\|_F} \quad (10)$$

where $\Delta_{\text{output}}^{\text{SLB}}$ is restricted to the last overlap/new frames—the portion that becomes the next chunk’s SLB. This measures sensitivity of the boundary output to boundary

input perturbations, which is the relevant quantity for SLB-mediated drift propagation. It does not capture drift arising from other channels (e.g., prompt conditioning or attention to non-overlap frames), so g_{SLB} is a lower bound on the true open-loop gain.

The `measure_g` config field existed from the start, but — caught only when this was checked directly rather than assumed — no code actually implemented Eq. (10) behind it; the flag was a documented no-op, hardcoded off with no node-level toggle. This has since been implemented: a second denoising pass with a perturbed overlap latent, holding the noise sample and every other conditioning input fixed, comparing the raw (pre-AdaIN/shrink) output tail against the unperturbed pass. It is verified at the tensor level — the perturbation is confined exactly to the overlap region, the norm-ratio formula recovers a known injected gain in a synthetic test — but **was not enabled in either experiment reported in Section 6**, both of which predate the implementation and ran with `measure_g=False` regardless; consequently no g_{SLB} value appears in any results table here, and the measurement itself is untested against a live transformer forward pass. The $g \approx 1.03\text{--}1.05$ figures in Section 7 are a *different*, informal proxy — the ratio of measured band-3 latent energy growth to its EMA reference — not an instance of Eq. (10). The two should not be conflated: the band-energy proxy reflects unshrunk spectral drift under the full correction stack as actually run, while g_{SLB} would isolate the transformer’s own sensitivity to a boundary perturbation, holding every correction fixed. Executing `measure_g=True` on a future run is the direct way to close this gap; it is listed as an open item in Section 8.

4 Algorithm

Algorithm 1 summarises the per-chunk CLSS procedure.

Algorithm 1 CLSS per-chunk generation

Require: Config: $\tau_c, \beta, \lambda, \gamma$; state: SLB, EMA, anchor bank

- 1: $L_{\text{ov}} \leftarrow \text{SLB.read}()$ ▷ may be empty for chunk 0
 - 2: $\tilde{L}_{\text{ov}} \leftarrow (1 - \tau_c) L_{\text{ov}} + \tau_c \varepsilon$ ▷ Eq. (1); Sect. 3.1
 - 3: $A \leftarrow \text{anchor_bank.retrieve}(\text{top-}m \text{ by cosine sim})$ ▷ Sect. 3.4
 - 4: $L_N \leftarrow \text{Denoise}(\tilde{L}_{\text{ov}}, A, \text{prompt}_N)$
 - 5: $L_N^{\text{new}} \leftarrow L_N[\text{overlap_lf} :]$ ▷ drop overlap region
 - 6: $L_N^{\text{new}} \leftarrow \text{AdaIN_blend}(L_N^{\text{new}}, \text{EMA}, \beta)$ ▷ Sect. 3.2
 - 7: $L_N^{\text{new}} \leftarrow \text{band_soft_shrink}(L_N^{\text{new}}, \text{band_EMA}, \gamma)$ ▷ Sect. 3.3
 - 8: $L_N^{\text{new}} \leftarrow \text{detail_anchor}(L_N^{\text{new}}, E_{\text{lo/hi}}^{\text{ref}})$ ▷ Eq. (7); Sect. 3.5
 - 9: $\text{EMA.update}(L_N^{\text{new}}, \lambda)$; $\text{band_EMA.update}(L_N^{\text{new}}, \lambda)$
 - 10: $\text{anchor_bank.update}(\text{last frame of } L_N^{\text{new}})$ ▷ Sect. 3.4
 - 11: $\text{SLB.push}(L_N^{\text{new}}[-\text{overlap_lf} :])$
 - 12: **return** L_N^{new}
-

5 Implementation

5.1 Memory Budget for 16 GB VRAM

LTX-2.3 22B in BF16 occupies ≈ 44 GB. Three complementary strategies fit it within 16 GB VRAM:

1. **GGUF 4-bit quantisation** (Q4_K_S): reduces the on-disk size to ≈ 11 GB. De-quantization to BF16 at load time requires ≈ 44 GB of pinned CPU RAM.
2. **CPU block-streaming**: all transformer blocks reside in pinned CPU RAM; only 2 blocks are resident on GPU at a time (≈ 5 GB model weights on GPU), with the next block pre-fetched asynchronously via a dedicated CUDA stream.
3. **Temporal chunking (CLSS)**: latent memory is $O(\text{overlap_lf})$ instead of $O(\text{total_frames})$. Processed chunk latents are immediately offloaded to CPU after each chunk.

Combined peak VRAM budget at 512×768 resolution:

Component	Peak VRAM
Model blocks ($2 \times$ BF16, streaming)	≈ 5 GB
Activations + latents (per chunk)	≈ 4 GB
VAE / text encoder (loaded/freed)	≈ 2 GB
Total peak	≈ 11 GB

5.2 Two-Stage Pipeline

Stage 1 (Open-resolution streaming). CLSS streaming generates the full video at a coarse latent resolution ($H_L \times W_L$ latent grid, typically 11×20 for 704×512 output). Each chunk runs the full 20–30-step LTX scheduler. CLSS corrections are applied between chunks as per Algorithm 1.

Stage 2 (Distilled-LoRA refinement). Stage 1 operates at 704×512 output resolution and 20 denoising steps, which is sufficient for temporal structure and scene content but produces visible quantisation artefacts and reduced texture detail, particularly after GGUF 4-bit dequantisation at the low-frequency latent scale. Stage 2 addresses this by upsampling and refining the output.

The Stage 1 output is spatially upsampled $2\times$ by `LTXVLatentUpsampler`, yielding a 22×40 latent grid corresponding to 1408×1024 pixel resolution. Stage 2 then refines the full video in temporal chunks using a 3-step distilled-LoRA schedule with initial noise level $\sigma_0 \approx 0.909$. Each chunk’s new frames are seeded from the upscaled Stage 1 latent with σ_0 -scaled noise:

$$x_{\text{start}} = \sigma_0 \varepsilon + (1 - \sigma_0) L_N^{\text{S1}} \quad (11)$$

This formulation anchors Stage 2 to the coarse temporal structure produced by Stage 1—preserving scene layout, motion trajectories, and chunk boundaries established by

CLSS—while allowing the 3-step refinement to add local high-frequency detail. Stage 1 alone could in principle be generated at $2\times$ resolution, but this would quadruple token count and exceed VRAM budget under CPU block-streaming.

Key differences from Stage 1:

- AdaIN and spectral-shrinkage corrections are *disabled*: Stage 2 refinement is structurally anchored to the S1 latent, so distributional drift correction is unnecessary.
- The detail anchor (Section 3.5) *does* run, per window, with the reference re-computed from that window’s upscaled Stage-1 slice rather than carried from a first-window EMA. The 3-step distilled refinement systematically adds spatial high-frequency energy relative to its Stage-1 seed, and without the anchor this excess compounds across the window chain as end-of-run softening of the refined output relative to Stage-1 structure. Per-window re-referencing means the correction cannot accumulate: each window is equalised to its own source.
- A global pre-generated noise field is sliced per chunk (`_SlicedNoise`) to prevent grain seams at chunk boundaries—a single coherent noise tensor ensures that the per-chunk noise samples tile seamlessly when concatenated.
- Audio is *frozen*: the Stage-1 audio latent is passed through unchanged (denoise mask 0) and only video tokens are refined, with the frozen audio serving as cross-modal context. Re-rendering audio in Stage-2 windows was measured to re-seam it at every window boundary; freezing keeps the audio timeline bit-identical to Stage 1 (`frozen_verify_sim=1.0` per window).

5.3 Audio Streaming

Audio and video share the same temporal chunk boundaries. Audio length accounting is anchored to *pixel time*: the causal video VAE maps a chunk of `new_lf` latent frames to $(\text{new_lf} - 1) \cdot 8 + 1$ pixel frames for the first chunk of a sequence and $\text{new_lf} \cdot 8$ pixel frames for every continuation chunk. The audio rate is taken from the chunk template, $r = \text{new_af} / ((\text{new_lf} - 1) \cdot 8 + 1)$ audio frames per pixel frame, and the SLB length follows the same rate:

$$\text{overlap_af} = \text{round}(\text{overlap_lf} \cdot 8 \cdot r) \quad (12)$$

With `overlap_lf = 8` and the 13-lf/102-af template this gives `overlap_af = 67`. (An earlier per-chunk accounting ignored the causal first-frame discount, undercounting every continuation chunk by $\approx 0.28\text{s}$ and desynchronising audio from video by $\approx 2\text{s}$ over seven chunks; the pixel-time ledger above eliminates the drift exactly.) The effective video overlap is additionally clamped to `new_lf`: the SLB can never carry more frames than a chunk produces, and unclamped values were measured to silently discard motion at every seam.

Continuity via reference audio, not a buffer. Following the reference pipeline, audio continuity is carried by *conditioning*, not by a seeded buffer: the overlap-time audio of each continuation chunk is regenerated at full denoise and dropped, and

the only carrier of audio history is a reference-audio window (`ref_audio` tokens at negative temporal RoPE) taken from the previous chunk’s pre-overlap tail. An audio analogue of the video SLB (seeding the overlap frames at τ_c) exists as an option and was measured to make things worse when enabled: the buffer freezes the previous chunk’s loudest tail into every chunk’s opening, compounding into high-frequency runaway ($1.02 \rightarrow 1.51$ over ten chunks) and raw-RMS runaway that the energy anchor then fights with growing corrections. The reference window is passed *clean and at full length* (default 67 af ≈ 2 s, decoupled from the video overlap; an implicit coupling to `overlap_af` starved it at small overlaps and measurably collapsed within-chunk audio coherence from 0.91 to 0.27). Perturbing this reference — blending in noise, shrinking its window over chunks, flattening its loudness envelope — was each tested live and none reduced the envelope repetition it was aimed at, while the energy-injecting variants added audible hiss; the clean full-length reference is the measured optimum (Section 7.6).

Reference-exact RMS anchoring. Un-anchored streaming audio is a divergent loop: runs were measured with RMS growing +67% and low-mid spectral bands tripling over ten chunks. A per-chunk anchor corrects the new audio’s global RMS toward an EMA target whose drift from the chunk-1 reference (computed onset-excluded) is hard-capped at $\pm 5\%$ (the same Δ_σ used for the video EMA), before the chunk’s tail is fed forward as the next reference window. Anchoring *before* feed-forward is what keeps the recursion safe at any sequence length. In the T2V run this anchor absorbs a raw loudness drift of $0.54 \rightarrow 0.91$ (+72%), holding output RMS at 0.81–0.84 against a chunk-1 reference of 0.853 (Section 6.2.3). On multi-scene runs the anchor is re-baselined at scene changes; this is measurably a ratchet (each re-baseline legitimises accumulated loudness creep: $0.599 \rightarrow 0.699 \rightarrow 0.823$ over three scenes in one run) and remains a known limitation.

Boundary smoothing. At each audio chunk boundary, a linear blend over ± 2 audio latent frames is applied in latent space before VAE decoding. This eliminates click artifacts that arise from the abrupt splice between consecutive chunks, which would otherwise be visible as discontinuities in the spectrogram.

5.4 ComfyUI Node Integration

Five custom nodes implement the full pipeline inside ComfyUI:

1. **CLSSConfig**: exposes the CLSS hyperparameters worth exposing ($\tau_c, \beta, \text{overlap}$); validated internals are fixed.
2. **CLSSScenePrompts**: multi-scene text input via `---` delimiters; Gemma-enhanced per scene. Output is a flat CONDITIONING list with one entry per scene, proportionally assigned to chunks. With one prompt per chunk this doubles as a per-chunk *object-identity* anchor (see Section 8).

3. **CLSSStreamingSampler**: Stage 1 streaming sampler. Accepts any ComfyUI GUIDER/SAMPLER/SIGMAS/NOISE/LATENT combination. Emits an unconditional settings dump, per-chunk telemetry, end-of-run trend summaries with per-chunk localisation, per-frame origin/layout drift event tables, and a phase-lock repetition detector.
4. **CLSSStage2**: Stage 2 refinement sampler with consistent per-chunk noise slicing, per-window detail anchoring, and frozen-audio passthrough (Section 5.2).
5. **CLSSAVGuiderV2**: reference-parity multi-modal guider (Section 5.5). Its predecessor (**CLSSAVGuider**, a `sampler_cfg_function` patch) is retained but inert on current ComfyUI and warns accordingly.

All stochastic inputs are seeded: Stage-1 video *and* audio noise are pre-generated as full-sequence fields sliced per chunk (distinct yet coherent noise per chunk, bit-reproducible runs), and every fallback noise path uses a generator derived deterministically from the seed and chunk position.

5.5 Multi-Modal Guidance and the Packed-Latent Pitfall

The reference pipeline guides each modality independently (**MultiModalGuider**):

$$\text{pred} = c + (s_{\text{cfg}} - 1)(c - u) + s_{\text{stg}}(c - p) + (s_{\text{mod}} - 1)(c - m) \quad (13)$$

per modality, followed by a rescale $\text{pred} *= r \sigma_c / \sigma_{\text{pred}} + (1 - r)$, where u is the text-unconditional prediction, m the modality-isolated prediction (both audio↔video cross-attentions skipped), and p the STG-perturbed prediction. The STG perturbation skips *self*-attention in designated blocks (block 28 for LTX-2.3) by replacing the attention output with the raw value projection ($\text{out} = W_V x$, gating and output projection unchanged) — a value-passthrough, not a zero-out. The reference defaults are video cfg 4–4.5, audio cfg 7, modality 3, STG 1 for both modalities, rescale 0.7 (four transformer passes per step). The working configuration runs $s_{\text{mod}} = 1$: the modality term of Eq. (13) vanishes and its pass is skipped (three passes per step). This is a measured choice, not an economy: raising modality guidance to the reference value 3 was tested live and made the audio audibly *worse* (hiss/drone; within-chunk audio coherence collapsed $0.68 \rightarrow 0.61$) while leaving the audio repetition it might have addressed unchanged — with Stage-2 audio frozen, the raw guided Stage-1 audio is the final audio, and additional guidance energy surfaces directly as audible high-frequency content.

The packed-latent representation. ComfyUI’s `CFGGuider.sample()` packs nested audio-video latents into a *single flat* $[B, 1, N_v + N_a]$ tensor before sampling and unpacks only after. A guider that tries to detect the audio-video case with `isinstance(x, NestedTensor)` inside `predict_noise` therefore never sees a nested tensor and falls back to plain shared CFG — and it does so silently, producing no error and no log line. The guider must instead capture the per-modality shapes in `sample()` *before* packing, detect the packed $[B, 1, N]$ representation in `predict_noise`, unpack each

pass output, apply Eq. (13) per modality, and repack. Two diagnostics guard against a silent regression: the guider announces the active branch and its pass count once per chunk (`passes/step=3` for the working configuration, 4 with modality guidance enabled), and it logs the STG difference norms so a zero-valued (inert) perturbation is visible immediately. A useful invariant follows from the pass count: because each guidance term costs a transformer pass, a guidance configuration that does not change per-step wall-clock time is not actually running — in an earlier 13-lf configuration the four-pass stack cost $\approx 17.3\text{s/it}$ against $\approx 8.0\text{s/it}$ for plain CFG. Two engineering rules generalise: guidance paths must announce which branch is active, loudly and unconditionally; and a guidance change that does not change step time did not happen.

6 Experiments

Two generation experiments share the same CLSS configuration, seed, and active guidance stack but differ in the first-frame conditioning: a text-to-video (T2V) run with no image prior (Section 6.2), and an image-to-video (I2V) run conditioned on a single guide frame ($1 \times 11 \times 20$ latent, Section 6.3). Both stream ten Stage-1 chunks (`new_lf = 16`, `overlap = 8`) to 160 latent frames ($\approx 53\text{s}$ at 24fps) and refine through the same four-window Stage 2 pass.

6.1 Shared Configuration

Table 1: Shared configuration. Guidance values are those actually active during sampling (verified by the guider’s per-chunk `passes/step` and difference-norm telemetry, Section 5.5). The Stage-2 geometry is selected automatically from the VRAM token budget.

Parameter	Value
Model	LTX-2.3 22B (Q4_K_S GGUF)
Resolution	704 × 512 (Stage 1), 1408 × 1024 (Stage 2)
τ_c	0.05
β	0.40
λ	0.10
Δ_σ	0.05
F_a	3 (auto, for 10 chunks)
γ	(0.0, 0.05, 0.2, 0.3)
Detail anchor	on (Stage 1 scene-first ref; Stage 2 per-window ref)
Video guidance	CFG 4.0, STG 1.0
Audio guidance	CFG 7.0, STG 1.0
Modality guidance	1.0 (disabled; Section 5.5)
STG block / rescale	28 / 0.70 (three passes/step)
S1 noise temporal-corr mix	0.30 (video only)
Audio continuity	<code>ref_audio</code> 67 af, clean; no audio SLB
Audio anchor	RMS-only, reference-exact ±5%
Stage 1 steps	20
Stage 2 steps	3 (distilled), audio frozen (mask 0)
VRAM (GPU)	15.61 GB (RTX-class)
T2V first-frame	Text only
I2V first-frame	Guide image, latent [1, 128, 1, 11, 20]
Stage 1 (both)	10 chunks, <code>new_lf</code> = 16, <code>overlap</code> = 8, 1 scene
Stage 2 (both)	4 windows, <code>fpc</code> = 40, <code>overlap</code> = 13 (auto)

6.2 Experiment 1: Text-to-Video (T2V)

No image is provided; Stage 1 chunk 1 begins from pure noise conditioned on the text prompt alone. The clip is a single prompted scene streamed as ten chunks (`new_lf` = 16, `overlap` = 8), assembling to 160 latent frames ($\approx$ 53s), under the identical seed and guidance stack as the I2V run (Table 1). The prompted content is highly dynamic: intra-chunk similarity (a chunk’s first vs. last new frame) sits near zero for most of the run (Table 3), so this run exercises the correction stack under continuous motion rather than a stable anchored scene.

6.2.1 Guidance Activity

Every chunk shows `passes/step=3` with non-zero STG difference norms on both modalities (Table 2); no step fell back to plain shared CFG. Chunk 1’s audio STG norm is the outlier (18.5 versus 5.4–12.2 thereafter): with no reference-audio window yet, the perturbed audio prediction diverges further from the conditional one. The same chunk-1 outlier appears in I2V on the *video* norm instead (Table 7), where the guide image is the unmatched conditioning.

Table 2: T2V Stage 1 guidance difference norms (step 1 of each chunk). Modality guidance is disabled ($s_{\text{mod}} = 1$), so only the STG terms are reported.

Chunk	STG (vid)	STG (aud)
1	84.1	18.5
2	88.6	12.2
3	98.4	6.2
4	140.3	9.6
5	112.9	8.0
6	158.5	5.4
7	221.8	10.6
8	127.3	9.0
9	167.8	8.8
10	131.0	9.9

6.2.2 Stage 1 Results

Table 3: T2V Stage 1 per-chunk video metrics (10 chunks).

Chunk	Boundary sim $\uparrow$	Intra sim	Identity sim	ρ_{loop}
1/10	N/A	0.1683	1.0000 (ref)	0.570
2/10	0.7930	0.1893	0.7930	0.570
3/10	0.3826	0.3825	0.7318	0.570
4/10	0.5515	0.0973	0.7656	0.570
5/10	0.7340	−0.0315	0.7340	0.570
6/10	0.8893	0.2079	0.6630	0.570
7/10	0.8283	0.2498	0.7099	0.570
8/10	0.9562	0.1992	0.9562	0.570
9/10	0.9715	0.1239	0.6967	0.570
10/10	0.9240	0.8520	0.9325	0.570

Without a guide image, video boundary continuity is far more variable than I2V’s: the early seams range down to 0.383 (chunk 3) while the last three seams sit at 0.92–0.97, matching I2V levels. The trajectory is the opposite of error accumulation — the

weakest hand-offs are *early*, while the content is still finding a stable configuration, and continuity *improves* monotonically from chunk 5 onward as the SLB context becomes self-consistent. Intra-chunk similarity near zero for most chunks (even negative at chunk 5) confirms sustained motion within every chunk; the jump to 0.852 at chunk 10 marks the content settling in the final seconds. Origin similarity to the scene reference is contained over the run ($0.142 \rightarrow 0.129$, range 0.091), and identity similarity to the anchor bank holds a 0.66–0.96 band with no monotonic decay. The largest origin-similarity drops still concentrate at the chunk 1→2 hand-off ($t = 5.3\text{--}7.3\text{ s}$, five of the six largest events, deepest $\Delta = -0.353$), while the largest coarse-layout drops are spread across chunks 1, 2, and 4 — consistent with a run whose content moves throughout rather than drifting from a fixed opening.

Table 4: T2V Stage 1 AdaIN corrections, band gains, and detail-anchor gains ($g_{\text{lo}}/g_{\text{hi}}$, Eq. (7); 0.9000 is the per-chunk floor cap).

Chunk	$\Delta\mu$	$\Delta\sigma$	g_1	g_2/g_3	g_{lo}	g_{hi}
1	+0.0000	+0.0000	1.0000	1.0000/1.0000	ref	ref
2	+0.0019	−0.0052	0.9962	0.9829/0.9892	1.0146	1.0083
3	−0.0085	−0.0431	0.9882	0.9549/0.9782	1.0499	0.9226
4	−0.0028	−0.0430	0.9902	0.9706/0.9876	1.0205	0.9326
5	−0.0048	−0.0405	0.9910	0.9679/0.9865	1.0305	0.9283
6	+0.0044	−0.0392	0.9917	0.9678/0.9927	1.0437	0.9238
7	+0.0055	−0.0811	0.9843	0.9446/0.9699	1.0309	0.9000
8	+0.0047	−0.0506	0.9875	0.9683/0.9894	1.0640	0.9000
9	+0.0037	−0.0494	0.9911	0.9757/0.9933	1.0280	0.9000
10	+0.0034	−0.0480	0.9906	0.9738/0.9932	1.0929	0.9000

Every T2V continuation chunk receives a *downward* std correction ($\Delta\sigma < 0$ throughout), in contrast to I2V’s near-neutral pattern (Table 9): without an image-conditioned first chunk to compress toward, each text-only chunk’s raw std runs hot relative to the EMA reference and AdaIN consistently pulls it back down. Spectral shrinkage is correspondingly more active than in I2V (band-2 gains down to 0.9446 vs. I2V’s ≥ 0.985). The detail anchor works visibly hard in this run: raw high-band energy overshoots its scene-first reference on every chunk from chunk 3 (the model keeps adding spatial detail energy under sustained motion), and from chunk 7 onward g_{hi} rides its 0.90 floor cap while g_{lo} boosts the low band — the net effect is a high-frequency *share* held at $0.512 \rightarrow 0.520$ over the full run (trend range +0.020), where the same one-sided shrinkage without the anchor produced a monotonic hf-share collapse (Section 3.5).

The anchor bank grows identically to I2V, from 1 to 5 entries at the same frame positions (bank_frame_ids = [15, 47, 63, 111, 159]). With new_lf = 16, chunk k ’s last frame lands at absolute index $16k - 1$: 15, 31, 47, 63, ... for $k = 1, 2, 3, \dots$ — so these five entries are chunks {1, 3, 4, 7, 10}, not chunks 2, 5, 6, 8, or 9. Four of the five are the forced-insertion schedule exactly (chunk_index mod $F_a = 0$ with $F_a = 3$: chunk 1

always seeds, then chunks 4, 7, 10). Chunk 3 is the one *organic*, threshold-triggered commit of the run: chunk 2’s similarity to the seed anchor was already below threshold (last_frame_max_sim = 0.4171, streak= 1), and chunk 3 pushed the streak to $P = 2$, committing independently of the forcing schedule. Because the forced positions are fixed by chunk index alone, and T2V and I2V share chunk boundaries and seed, the two runs’ forced entries coincide exactly despite entirely different video content; only an organic (content-triggered) entry could differ between them, and none did here (I2V’s chunk 2 sits at 0.4183 against the same threshold — the organic commit lands at the same chunk in both runs).

6.2.3 Stage 1 Audio

Table 5: T2V Stage 1 per-chunk audio metrics. RMS raw is the chunk’s pre-anchor loudness; RMS out is after the reference-exact anchor (chunk-1 onset-excluded reference 0.853).

Chunk	WC start	WC end	env_corr	RMS raw	RMS out	Bnd sim
1	0.729	0.565	—	0.853	(ref)	N/A
2	0.457	0.252	0.247	0.542	0.822	0.3855
3	0.710	0.342	0.733	0.675	0.810	0.3445
4	0.795	0.505	0.911	0.772	0.810	0.4132
5	0.798	0.566	0.960	0.805	0.810	0.5519
6	0.877	0.689	0.790	0.810	0.810	0.5969
7	0.864	0.718	0.829	0.847	0.814	0.8174
8	0.842	0.671	0.873	0.884	0.821	0.5013
9	0.851	0.748	0.871	0.911	0.830	0.5898
10	0.830	0.676	0.758	0.909	0.838	0.5282

The reference-exact RMS anchor earns its name in this run. Raw per-chunk loudness collapses at chunk 2 (0.542 against a chunk-1 reference of 0.853, corrected with gain 1.52) and then creeps monotonically to 0.911 by chunk 9 (+68% from its own floor); the anchored output holds 0.81–0.84 throughout, within $\pm 5\%$ of the reference. Without the anchor this run would have both a loudness dropout at the first hand-off and a +72% end-to-end drift.

The loudness-envelope correlation climbs from 0.247 at chunk 2 into a sustained 0.76–0.96 band from chunk 4 onward — the same loudness gesture recurring each chunk, the envelope-repetition failure mode analysed in Section 7.6. The energy-peak detector reads looser than the correlation suggests (mode covers 3/10 chunks within ± 1 ; peaks at frames 26–34 early, 113 late), i.e. the repeated gesture drifts in phase across the run rather than locking to one frame. Within-chunk coherence is lower than I2V’s late-run values and boundary similarity is moderate (0.34–0.82), consistent with the dynamic content: each chunk’s audio tracks its own visual motion first and its predecessor second.

6.2.4 Stage 2 Results

Table 6: T2V Stage 2 per-window metrics (4 windows). g_{lo}/g_{hi} are the per-window detail-anchor gains (reference: the window’s own upscaled Stage-1 slice); Aud_bnd is the frozen Stage-1 audio’s own continuity at the window seam (Stage 2 never re-rolls audio).

Window	Bnd sim $\uparrow$	Intra sim	Fid first	Fid last	Aud_bnd	g_{lo}/g_{hi}
1/4	N/A	−0.1514	0.8906	0.9466	N/A	1.054/0.900
2/4	0.7833	0.7804	0.9514	0.9460	0.6759	1.032/0.900
3/4	0.8214	0.8260	0.9264	0.9218	0.8269	1.006/0.900
4/4	0.7782	0.7803	0.8343	0.8448	0.8956	1.011/0.900

Window 1’s intra similarity is *negative* (−0.151), unlike I2V’s positive 0.600 for the same window: the dynamic T2V opening (Table 3, chunk-1 intra 0.168) carries into Stage 2 as a first window whose start and end frames barely correlate, whereas I2V’s guide image keeps its first window internally coherent. The per-window detail anchor engages identically at every window: the 3-step distilled refinement adds spatial high-frequency energy over its Stage-1 seed at every window (pre-anchor hf share 0.48–0.56 against Stage-1 references of 0.35–0.42), and g_{hi} rides its 0.90 floor cap in all four windows, pulling the excess back toward the Stage-1 reference. Because the reference is recomputed per window, the correction cannot compound: fidelity to Stage-1 structure holds 0.92–0.95 through window 3, easing to 0.84 only in the final window. The audio row measures the *frozen* Stage-1 audio’s own continuity at the three window seams (0.676/0.827/0.896) — Stage 2 introduces no audio seams of its own (frozen_verify_sim= 1.0 at every window).

6.3 Experiment 2: Image-to-Video (I2V)

A single guide image is encoded to a latent of shape [1, 128, 1, 11, 20] and provided as first-frame conditioning to Stage 1 chunk 1 (guide adherence 1.0000 measured at the placed frame). The clip is a single prompted scene streamed as ten chunks (new_lf = 16, overlap = 8), assembling to 160 latent frames ($\approx$ 53s). All CLSS hyperparameters and the guidance stack are those of Table 1.

6.3.1 Guidance Activity

The guider emits **passes/step=3** and its STG difference norms once per chunk (Table 7); both modalities stay well away from zero on every chunk, so no step fell back to plain shared CFG. Chunk 1 is the outlier on the *video* STG norm (200.7 versus 76.9–172.5 thereafter): the guide image is a hard constraint the perturbed prediction cannot honour, so the perturbation response is largest exactly where the conditioning is strongest.

Table 7: I2V Stage 1 guidance difference norms (step 1 of each chunk). STG measures the self-attention-skip perturbation; non-zero $\Rightarrow$ guidance active. Modality guidance is disabled ($s_{\text{mod}} = 1$).

Chunk	STG (vid)	STG (aud)
1	200.7	10.5
2	85.3	7.3
3	132.3	5.7
4	93.6	6.5
5	96.6	4.1
6	88.7	5.7
7	76.9	7.2
8	91.1	5.3
9	100.6	7.4
10	172.5	8.7

6.3.2 Stage 1 Results

Table 8: I2V Stage 1 per-chunk video metrics (10 chunks). Boundary_sim compares the last overlap frame of the previous chunk against the first new frame; identity_sim compares the chunk against its nearest anchor-bank entry.

Chunk	Boundary sim $\uparrow$	Intra sim $\uparrow$	Identity sim	ρ_{loop}
1/10	N/A	0.5006	1.0000 (ref)	0.570
2/10	0.9245	0.5422	0.9245	0.570
3/10	0.9389	0.8963	0.8963	0.570
4/10	0.9366	0.9088	0.9366	0.570
5/10	0.9188	0.9307	0.9188	0.570
6/10	0.9415	0.9290	0.9101	0.570
7/10	0.9498	0.9431	0.9431	0.570
8/10	0.9654	0.9378	0.9654	0.570
9/10	0.9622	0.9377	0.9670	0.570
10/10	0.9741	0.7601	0.9147	0.570

Video boundary continuity holds between 0.919 and 0.974 across all nine chunk transitions and *rises* over the run — the last three seams are the strongest three. The floor sits early (chunk 5, 0.9188; chunk 2, the first SLB-conditioned hand-off, is barely above it at 0.9245), so the corrected loop does not compound error over this horizon. The largest structural events of the whole run still cluster at the chunk 1 $\rightarrow$ 2 seam ($t \approx 5.3\text{--}9.3\text{s}$): five of the six largest per-frame origin-similarity drops land there (deepest $\Delta = -0.219$), the transition out of the guide-image-locked opening. Global origin similarity to the scene reference is contained over the full run ($0.332 \rightarrow 0.262$, range 0.079), in contrast to the monotonic walk-away drift seen at wider overlaps

in earlier open-loop testing. Identity similarity to the anchor bank is essentially flat ($0.925 \rightarrow 0.915$, floor 0.896 at chunk 3), and intra-chunk similarity *grows* from 0.54 to a 0.93–0.94 plateau as the scene settles — a stable anchored scene, the complementary regime to T2V’s sustained motion.

Table 9: I2V Stage 1 AdaIN corrections, band gains, and detail-anchor gains (Band 0 is the DC band, $\gamma_0 = 0$, gain $\equiv 1$, omitted).

Chunk	$\Delta\mu$	$\Delta\sigma$	g_1	g_2/g_3	g_{lo}	g_{hi}
1	+0.0000	+0.0000	1.0000	1.0000/1.0000	ref	ref
2	−0.0027	−0.0115	0.9952	0.9863/0.9964	1.0130	0.9798
3	+0.0005	−0.0062	0.9977	0.9917/0.9976	1.0131	0.9857
4	−0.0017	−0.0118	0.9965	0.9906/0.9976	1.0091	0.9783
5	+0.0005	−0.0103	0.9963	0.9912/0.9974	1.0064	0.9824
6	+0.0006	−0.0136	0.9965	0.9902/0.9974	0.9971	0.9813
7	+0.0016	−0.0145	0.9964	0.9931/0.9980	0.9865	0.9868
8	+0.0027	−0.0140	0.9963	0.9926/0.9968	0.9923	0.9807
9	+0.0005	−0.0080	0.9971	0.9934/0.9968	1.0051	0.9807
10	−0.0020	−0.0139	0.9953	0.9854/0.9916	1.0223	0.9591

On this stable anchored scene the correction stack idles, as designed: AdaIN deltas stay within $|\Delta\mu| \leq 0.003$, $|\Delta\sigma| \leq 0.015$, every band gain is ≥ 0.985 , and the detail-anchor gains never stray more than 2.2% (low band) / 4.1% (high band) from unity — an every-chunk trim, not a fight. The result is the flattest telemetry of either run: high-frequency share exactly flat ($0.510 \rightarrow 0.510$, trend range 0.000), std stable ($0.959 \rightarrow 0.962$). Contrast T2V (Table 4), where the same stack works at its caps every chunk; the corrections scale with measured excess rather than applying a fixed schedule.

The anchor bank grows from 1 to 5 entries over the run at chunks $\{1, 3, 4, 7, 10\}$ (bank_frame_ids = [15, 47, 63, 111, 159]) — the same forced-insertion-plus-one-organic-commit pattern derived in detail for T2V (Section 6.2), since both runs share chunk boundaries and seed.

The ComfyUI sampler uses the bank for identity telemetry only: the retrieval-injection conditioning path of Section 3.4 is present in the reference library but not wired into this sampler, so the bank measures identity drift without correcting it. Two conditioning wirings have been tried and both failed (Section 8): the reference library’s own `VideoConditionByKeyframeIndex` appends a separate token block and reproduces the audio corruption documented for the i2v guide path (Section 5.5); an in-place low-strength replacement at the first new frame instead produced visible content morphing, because anchor retrieval returns the *least*-similar non-redundant anchor and so drags each chunk back toward content the scene has already moved past. No safe anchor conditioning mechanism is currently known; the bank remains diagnostic-only.

6.3.3 Stage 1 Audio

Table 10: I2V Stage 1 per-chunk audio metrics. Within-chunk similarity is reported at the start and end of each chunk; `env_corr` is the loudness-envelope correlation with the previous chunk (> 0.7 flags a repeating gesture); RMS raw/out are pre-/post-anchor (chunk-1 onset-excluded reference 0.532).

Chunk	WC start	WC end	env_corr	RMS raw	RMS out	Bnd sim
1	0.361	0.391	—	0.529	(ref)	N/A
2	0.384	0.652	0.402	0.498	0.528	0.0724
3	0.663	0.699	0.768	0.556	0.531	0.2981
4	0.634	0.754	0.844	0.559	0.534	0.4690
5	0.747	0.840	0.769	0.571	0.538	0.3455
6	0.580	0.775	0.587	0.525	0.536	0.5346
7	0.741	0.795	0.614	0.551	0.538	0.4699
8	0.638	0.737	0.599	0.537	0.538	0.4339
9	0.478	0.679	0.442	0.542	0.538	0.2670
10	0.457	0.656	0.463	0.546	0.539	0.4395

Loudness is intrinsically stable in this run: raw per-chunk RMS never leaves 0.50–0.57 and the anchor’s corrections never exceed 6.1% — the anchor trims rather than rescues (contrast T2V, Section 6.2.3). The loudness-envelope correlation traces an arc: it climbs from 0.402 at chunk 2 to a peak of 0.844 at chunk 4 — crossing the 0.7 repetition flag for chunks 3–5 — and then *decays* to 0.44–0.46 over the final two chunks; the energy-peak positions scatter correspondingly (frames 26, 3, 30, 31, 92, 35, 92, 126, 92, 10; mode covers only 3/10 within ± 1). Under the clean full-length reference window the correlation does not lock permanently, but the repeating-gesture signature remains the flagged drift of the trend summary in both runs, and per-chunk repetition remains audible in listening across every configuration tested — its structural character and the interventions that failed against it are analysed in Section 7.6. Audio boundary similarity is low in absolute terms (0.07–0.53): with the audio SLB off, the seam frame is not pinned, and continuity across the cut rests on the reference-audio conditioning alone.

6.3.4 Stage 2 Results

Stage 2 refines the $2\times$ -upsampled latent in four overlapping windows ($\text{fpc} = 40$, $\text{overlap} = 13$, auto-selected from the 42k-token VRAM budget) against frozen Stage-1 audio (Section 5.2).

Table 11: I2V Stage 2 per-window metrics (4 windows). Fid is video fidelity to the Stage-1 latent (first / last frame of the window); Aud_bnd is the frozen Stage-1 audio’s own continuity at the window seam; g_{lo}/g_{hi} are the per-window detail-anchor gains.

Window	Bnd sim $\uparrow$	Intra sim	Fid first	Fid last	Aud_bnd	g_{lo}/g_{hi}
1/4	N/A	0.5998	0.9870	0.9689	N/A	1.039/0.900
2/4	0.9281	0.9569	0.9733	0.9419	0.5499	1.038/0.900
3/4	0.7768	0.7820	0.9086	0.9296	0.8365	1.022/0.900
4/4	0.8802	0.8067	0.9200	0.9402	0.7136	1.028/0.900

End-of-window fidelity to the Stage-1 latent holds 0.93–0.97 across all four windows and *rises* over the final two (0.9296 $\rightarrow$ 0.9402) — no end-of-run erosion. As in T2V, the distilled refinement adds spatial high-frequency energy at every window (pre-anchor hf share 0.45–0.46 against Stage-1 references of 0.35–0.38) and the per-window detail anchor pulls it back with g_{hi} at the 0.90 floor cap throughout; because each window is referenced to its own Stage-1 slice, the trim cannot accumulate into progressive softening. Video seam continuity ranges 0.78–0.93, weakest at the window-2 $\rightarrow$ 3 seam ($t \approx 26.7$ s). The audio timeline is bit-identical to Stage 1 (`frozen_verify_sim`= 1.0 per window); the Aud_bnd column re-measures the Stage-1 audio’s own chunk continuity where the video windows happen to cut (0.55–0.84). The assembled refined latents are [1, 128, 160, 22, 40] (video, std 0.912) and [1, 8, 1333, 16] (audio, frozen), with no NaN/Inf.

6.4 Computational Overhead

Table 12: Wall-clock timing for the I2V run on a 16 GB GPU (RTX-class, CPU block-streaming); the T2V run’s per-chunk/per-window timings match to within a few percent throughout (see text) and are omitted for brevity. Stage 1 runs three transformer passes per step (split CFG, STG); Stage 2 runs a single video refinement pass per window (audio frozen).

Stage / pass	Chunk	Steps	s/it	Time	Tokens
S1	1/10	20	18.3	$\approx$ 6.1 min	—
S1	2–10/10	20	24.2–24.4	$\approx$ 8.1 min each	—
S2 video	1/4	3	68.5	$\approx$ 3.4 min	35,200
S2 video	2–4/4	3	102–103	$\approx$ 5.1 min each	46,640

Total generation time: Stage 1 $\approx$ 79 min + Stage 2 $\approx$ 19 min $\approx$ **98 min** for a $\approx$ 53-second clip. The dominant cost driver is the three-pass guidance in Stage 1 — each denoising step runs three transformer passes (conditional, unconditional, STG-perturbed) rather than one. Chunk 1 is cheaper than the continuations (18.3 vs. 24.2–24.4 s/it) because it carries no 8-frame overlap context (16 vs. 24 latent frames in the

window). In Stage 2 the per-window cost rises $\approx 50\%$ after window 1 ($68.5 \rightarrow 102\text{--}103\text{s/it}$) against a token-count rise of $\approx 32\%$: as the run proceeds an increasing share of the model is pushed to CPU (offloaded weights grow from $\approx 5.7\text{GB}$ to $\approx 7.8\text{GB}$ across the pass), so the slowdown is disproportionate to the token increase and is an offload-pressure effect rather than a compute-scaling one.

The T2V run costs the same as I2V to within rounding: per-chunk and per-window s/it match to a few percent at every position (e.g. Stage-1 chunk 1: 18.2 vs. 18.3s/it ; Stage-2 window 3: 105.1 vs. 102.8s/it), confirming that the guide image changes neither the token count nor the compute cost — only the conditioning. The Stage-2 offload-pressure slowdown reproduces at the same windows in both runs, supporting that it is a property of the refinement pipeline rather than of run content.

7 Discussion

7.1 Effect of Chunk Length and Content Regime

Both runs used `new_lf` = 16 ($\approx 5.3\text{s}$ per chunk), within the 21-frame safe limit established in earlier open-loop testing, streamed as ten chunks under the active guidance stack. In neither run does boundary similarity degrade with chunk index — in both it *improves*: I2V’s three strongest seams are its last three ($0.9622\text{--}0.9741$), and T2V climbs from an early floor of 0.383 (chunk 3) to $0.92\text{--}0.97$ over the final three seams. The corrected loop does not compound error over this horizon in either conditioning mode; where continuity is weak, it is weak *early*, while the content is still finding its configuration. The two runs exercise opposite content regimes — I2V a stable anchored scene (intra-chunk similarity plateauing at $0.93\text{--}0.94$), T2V sustained motion (intra near zero for most chunks) — and the correction stack scales with the measured excess: near-idle on I2V (all band gains ≥ 0.985 , detail-anchor gains within $\pm 2\text{--}4\%$ of unity) versus working at its caps on T2V (band-2 gains to 0.9446 , g_{hi} pinned at the 0.90 floor for the last four chunks).

7.2 Identity Tracking and Boundary Self-Consistency

Chunk 2’s `identity_sim` (0.9245) equals its `boundary_sim` exactly. This is not a coincidence but a *self-consistency check*: the anchor was seeded from chunk 1’s last frame, and the SLB passes that same frame as the first latent frame of chunk 2’s conditioning, so the anchor query and the boundary query compare the same source frame (chunk 1 frame 15) against the same target frame (chunk 2 frame 0). Both metrics agreeing validates that the anchor bank and the SLB operate on the same reference, and that the τ_c re-noising at the boundary introduces only minimal distributional shift. For later chunks the bank holds multiple entries and the identity query selects a different anchor than the boundary, so the two metrics diverge, as expected (e.g. I2V chunk 6: identity 0.910 vs boundary 0.942). The same self-consistency check holds in the unrelated T2V run at the same chunk (identity 0.7930 = boundary 0.7930 , Table 3), confirming the mechanism is a property of the anchor/SLB bookkeeping rather than

of any particular content.

7.3 Audio Coherence and the Role of Visual Context

A natural hypothesis is that a guide image shapes streaming audio behaviour: in LTX-2.3’s joint audio-video transformer, video and audio tokens share self-attention layers, so a strong single-frame visual prior could constrain audio generation implicitly through cross-modal attention — for better (a coherence anchor) or worse (a repetition template). The matched pair does not support the hypothesis in either direction. Chunk-1 audio coherence is *higher* without the guide image (T2V 0.729 $\rightarrow$ 0.565 vs. I2V 0.361 $\rightarrow$ 0.391), and the envelope-repetition signature is *stronger* without it too (T2V sustains `env_corr` 0.76–0.96 from chunk 4; I2V peaks at 0.844 and decays to 0.46). Both differences track the prompted *sound content* — T2V’s loud continuous ambience (RMS reference 0.853) naturally self-correlates within and across chunks, while I2V’s quieter, more varied scene audio (RMS reference 0.532) does not — rather than the presence or absence of a visual prior. The practical conclusion is negative but useful: visual conditioning is not a lever on streaming audio behaviour at this scale, and the audio failure modes of Section 7.6 must be addressed in the audio path itself.

7.4 Stability Margin

With $\rho_{\text{loop}} = 0.570$, the CLSS feedback loop provides a $\approx 43\%$ gain reduction per chunk before accounting for the model’s open-loop gain g . In the I2V run the spectral-shrinkage correction was light throughout — band-2/3 gains of 0.985–0.998, a $\approx 1\text{--}3\%$ high-frequency energy excess over the EMA reference. Taking the typical $\approx 1\text{--}3\%$ per-chunk growth as a rough estimate of the open-loop gain in the spectral channel gives $g \approx 1.02$ and $\rho_{\text{closed}} \approx 0.57 \times 1.02 \approx 0.58$, comfortably below the stability boundary of 1. The T2V run runs hotter on this channel: band-2 gains fall to 0.945–0.975 for most chunks, a $\approx 4\text{--}6\%$ typical excess, giving $g \approx 1.05$ and $\rho_{\text{closed}} \approx 0.57 \times 1.05 \approx 0.60$ — still comfortably stable, and close enough to the I2V estimate that the margin appears to be a property of the correction stack rather than of the conditioning mode. The raw audio, however, shows a spectral drift the correction stack does not reach: the Stage-1 trend summary’s `aud_hf` statistic (a ratio of high-band to reference energy, not covered by the video-side machinery of Sections 3.3 and 3.5) grows in *both* runs — 0.42 $\rightarrow$ 1.33 in T2V (flagged as drift) and 0.87 $\rightarrow$ 1.28 in I2V — with T2V’s top frequency bins individually reaching 1.2–1.7 $\times$ the chunk-1 reference by the final chunks. This is audio-domain high-frequency energy growth with no video-channel analogue: the video hf channel is closed by the detail anchor, while no current correction targets the audio frequency distribution (the RMS anchor constrains only global loudness).

7.5 Stage-2 Identity Metric Ambiguity

Stage 2 lacks an anchor bank; its `identity_sim` reduces to comparing each window against the fixed first-window reference. On a single moving scene this conflates ordinary content motion with drift: in I2V, where the reference is the guide-image region, windows 2–4 sit at a 0.42–0.56 plateau, reflecting that the video has moved away from its opening frames, not instability. In T2V, where window 1 has no anchor to begin with, the same comparison instead oscillates slightly negative (-0.24 to -0.15 across windows 2–4) — consistent with window 1 itself being internally uncorrelated (Table 6, intra -0.151), so there is no stable reference for later windows to be “close to” or “far from” in the first place. Adding a scene-aware anchor bank to Stage 2 would make this metric interpretable in either conditioning mode; as reported it should be read only as “distance from the opening,” not as a drift signal.

7.6 Measured Failure Modes at Longer Horizons

Per-frame telemetry across extended runs — including the ten-chunk, ≈ 53 -second T2V and I2V runs reported above and earlier 13-lf and 31-lf runs — exposes two repetition mechanisms with distinct signatures, plus the Stage-2 audio design consequence.

Loudness-envelope repetition is structural. Streaming Stage-1 audio repeats a per-chunk loudness gesture: `env_corr` crosses the 0.7 repetition flag in both reported runs (sustained 0.76–0.96 in T2V; an arc to 0.844 in I2V), it is the flagged drift of both trend summaries, and the repetition is audible in listening across every configuration tested. Every intervention available at the conditioning and guidance level was tested live against it, and *all failed*: enabling the audio SLB (continuity carrier — repetition unchanged, plus high-frequency runaway $1.02 \rightarrow 1.51$ as the buffer froze each chunk’s loud tail forward); raising modality guidance to the reference value (repetition unchanged, plus audible hiss/drone as guidance energy rose); and perturbing the reference-audio window itself — blending in noise, decaying its length over chunks, flattening its loudness envelope (repetition unchanged in each case, with the energy-injecting variants adding hiss). The conclusion is that the mechanism is not in the conditioning at all: every chunk denoises against the same static text prompt, and the model re-renders its canonical loudness arc for that prompt each time — continuity conditioning modulates the details, guidance modulates the energy, and neither can remove the arc. Consistent with this account, the signature’s *strength* tracks the prompted sound content (strongest for T2V’s continuous ambience), not any conditioning knob. The only structurally sound fix is generating or re-rendering the audio in a single non-chunked pass, which for joint AV denoising is blocked by the video context’s token count at full length ($\approx 140k$ tokens against a 42k budget); it remains future work.

Chunk-relative phase lock is broken; the gesture remains. An earlier symptom of the same mechanism was a hard *phase* lock: the audio energy peak at 94% of

the window for eight consecutive 13-lf chunks. In the reported runs the phase-lock detector reads unlocked in both modes (peak mode covers only 3/10 chunks within ± 1 ; I2V peaks scatter across frames 3–126) — the repeated gesture now drifts in phase across chunks rather than pinning to one relative frame. Envelope correlation and the ear, however, still register the repetition: phase scatter is an improvement in the artefact’s regularity, not its presence.

A ~ 4 -second video layout oscillation. Coarse scene layout departs from the scene reference and partially returns with a ≈ 4 s wall-clock period that is *invariant to chunk length*: at 13-lf (4.3 s) chunks the dip lands at new-frame 5–7 in every chunk (appearing seam-locked), while at 31-lf (9.9 s) chunks it recurs within the chunk at the same seconds-period. Context strength and context length (overlap 3/5/8) were tested and neither removes it; the oscillation appears intrinsic to the model’s trained temporal horizon. In the reported 16-lf runs the layout-argmin detector reads unlocked in both modes (I2V mode 12 covering 5/10 chunks within ± 1 , T2V mode 2 covering 2/10) — at overlap 8 the strict per-chunk swing-and-return does not phase-lock to the seam grid, though global layout still wanders (largest per-frame layout drops of -0.91 in I2V and -0.66 in T2V, spread across the first half of each run rather than concentrated at one seam).

Stage-2 audio is frozen by design. The freeze (Section 5.2) is the outcome of a measured escalation. With per-window audio re-rendering at high denoise (σ -scale 1.0), $\approx 45\%$ of the audio is re-rolled per window and the re-rolls disagree at every window boundary (boundary cosine 0.32–0.42) *regardless* of the audio SLB length — the SLB pins the seam frame, not the style of the re-roll. At denoise 0.4 the pass polishes instead (seed similarity 0.79–0.81) yet seams still disagreed at 0.59–0.72: polishing narrows the disagreement but cannot close it while windows are denoised independently. Freezing (mask 0) closes it exactly — Stage 2 introduces no audio seams at all, and the only audio discontinuities in the final output are Stage 1’s own chunk boundaries. Rule: audio is either generated in one window or not re-rolled at all; any multi-window audio denoise re-seams the timeline it was meant to improve.

Unbounded audio high-frequency growth. The trend-summary’s `aud_hf` statistic (high-band audio energy relative to the chunk-1 reference) grows in both reported runs: $0.42 \rightarrow 1.33$ in T2V — flagged as drift, with top frequency bins individually reaching $1.2\text{--}1.7\times$ reference by the final chunks — and $0.87 \rightarrow 1.28$ in I2V. No current correction targets this channel: the video-side spectral shrinkage (Section 3.3) and detail anchor (Section 3.5) operate on the video latent only, and the audio-side RMS anchor (Section 5.3) constrains global loudness but not the frequency distribution. An audio-side two-band anchor, mirroring Section 3.5 but applied to the audio latent’s frequency axis, is the natural next mechanism to try; an earlier attempt at audio high-band shrinkage was reverted for audible artefacts, so any such correction must be validated by ear, not by the statistic it improves.

8 Limitations

Single hyperparameter configuration, single seed. The reported T2V and I2V results share one set of hyperparameters (new_lf = 16, overlap = 8, 10 chunks, 1 scene), one seed, and one hardware configuration (16 GB VRAM, RTX-class); they differ only in the first-frame conditioning, by design, to isolate that variable. No baseline comparison against open-loop streaming is reported, and no ablation of the individual CLSS components (re-noising alone, AdaIN alone, shrinkage alone, anchor bank alone) or of the individual guidance terms (split CFG, modality, STG) has been performed. The paper therefore cannot quantify the marginal contribution of each correction or guidance term, or demonstrate improvement over the open-loop baseline from the same metrics, and every single-run comparison in Section 7 (e.g. the phase-lock/guide-image association) is a $n = 1$ -per-arm observation, not a statistically powered result. These comparisons are left to future work.

Stability metrics vs. perceptual quality. The reported metrics—boundary cosine similarity, intra-chunk cosine similarity, audio within-chunk segment similarity—measure the *stability* of the latent representation across chunks. They do not assess the perceptual quality of the generated content, prompt adherence, or realism. Standard quality metrics (FID, IS, CLIP similarity, human evaluation) would be required to characterise the generated video as a content generation system. High boundary similarity is a necessary but not sufficient condition for high video quality.

Object-level identity is unobserved. Every reported metric is computed from mean-pooled channel features or low-band spatial maps and is therefore *object-blind*: a vehicle or person can be replaced by a different instance across a chunk boundary while boundary and identity cosines remain above 0.95. This failure class was observed perceptually and is invisible to the telemetry. High-similarity readings must not be interpreted as object persistence. The available mitigation is textual: one prompt per chunk via `CLSSScenePrompts`, re-describing the persistent objects explicitly. Object-level tracking metrics (detection/re-identification on decoded frames) are future work.

Stability model assumptions; g_{SLB} not yet measured. The linear gain model in Section 3.6 assumes that g is approximately constant and that the AdaIN and re-noising corrections act as independent multiplicative attenuators. Both assumptions simplify the analysis: the actual closed-loop dynamics involve higher-order interactions between corrections, guidance, and scene content. A more rigorous treatment (e.g., Lyapunov stability for the latent iteration, or a Monte Carlo sweep over perturbation seeds and chunk positions) would require characterising the denoiser’s Jacobian across a distribution of latent inputs, and is left to future work. Concretely: the g_{SLB} perturbation protocol (Eq. 10) is implemented and verified at the tensor level — the perturbation is confined exactly to the overlap region, and the norm-ratio formula recovers a known injected gain in a synthetic test — but it doubles per-chunk compute, so it is compiled off in the node (no exposed toggle) and has not been exercised

against a live transformer. Neither reported experiment used it: the $g \approx 1.02\text{--}1.05$ figures in Section 7 remain an informal band-energy proxy, not this measurement. Re-enabling it in code for a dedicated measurement run is the direct next step toward an empirically grounded, rather than assumed, value of g .

No working anchor-bank conditioning. The anchor bank tracks long-range identity but does not condition generation: both wirings attempted to date have failed. The reference library’s append-style `VideoConditionByKeyframeIndex` reproduces the AV audio corruption already diagnosed for the i2v guide path (Section 5.5). A low-strength in-place replacement at the first new frame — verified in isolation to place the anchor correctly and leave the SLB region untouched — was exercised in a full run and produced visible content *morphing*: it injected a past keyframe into every chunk, and because retrieval returns the least-similar non-redundant anchor, it dragged each chunk back toward content the scene had already moved past (one chunk was pulled to an earlier anchor than a previous chunk had used). It was reverted. Conditioning the joint audio-video model on a temporally-displaced keyframe without either corrupting audio (append) or morphing video (in-place replace) is an unsolved design problem; the ~ 4 -second video layout oscillation (Section 7.6) that motivated the attempt remains unaddressed.

Hardware specificity. The GGUF 4-bit quantisation and CPU block-streaming configuration was designed for a specific VRAM budget (16 GB). Performance, latency, and numerical behaviour may differ on GPUs with more VRAM (where block-streaming is unnecessary) or with different quantisation schemes (e.g., Q8 or FP8).

9 Related Work

Long video generation. StreamingT2V [3] and FIFO-Diffusion [4] generate long videos by sliding-window denoising. Gen-L-Video [5] uses temporal attention patching. FreeNoise [1] reschedules the initial noise (window-based fusion of a shuffled, reused noise sequence) to extend a pretrained model’s temporal attention window without retraining or sliding-window denoising at all — a tuning-free approach to the same underlying problem (fixed training-time context length) that CLSS addresses by chunking with closed-loop correction instead. VidToMe [2] merges self-attention tokens across frames to improve temporal consistency in zero-shot video editing, operating within a single generation pass rather than across autoregressively-generated chunks. CLSS differs from all of these by applying closed-loop corrections specifically designed for the latent-streaming regime of large audio-video models, where each chunk is a genuinely separate denoising process conditioned on the previous chunk’s output — the exposure-bias setting FreeNoise and VidToMe do not target.

Exposure bias in diffusion. The exposure bias problem in autoregressive models is well-studied [6]. For diffusion models, Ning et al. [7] and Salimans & Ho [8] address

related distributional mismatch issues during training, but inference-time corrections for streaming have received less attention.

Statistical normalisation for consistency. Adaptive Instance Normalisation (AdaIN) [9] was introduced for style transfer, where the goal is to match the mean and standard deviation of a content feature map to those of a style reference. EMA-based statistical normalisation across video frames has also appeared in temporal consistency methods. In CLSS, AdaIN is repurposed as a *drift correction* applied to denoised latents between chunks, using the cross-chunk EMA as the style reference. The combination of EMA tracking with AdaIN blending (rather than full replacement) is motivated by the streaming setting, where the reference must itself evolve to allow intentional scene changes.

Frequency-domain video editing. Yang et al. [10] and Ceylan et al. [11] use frequency-domain constraints for temporal consistency in video editing. CLSS applies frequency shrinkage in the *latent* domain to a streaming generation setting, targeting energy accumulation rather than pixel-domain flicker.

10 Conclusion

We presented CLSS, a five-component closed-loop streaming synthesis framework that enables stable long audio-video generation with LTX-2.3 22B on consumer 16 GB VRAM hardware. The corrections — calibrated context re-noising, EMA-tracked AdaIN, frequency-band soft shrinkage, a dynamic anchor bank, and a two-band spatial detail anchor — achieve $\rho_{\text{loop}} = 0.570$; driven by a reference-parity multi-modal guider (split per-modality CFG and STG) that correctly handles ComfyUI’s packed audio-video latent representation, the pipeline held Stage 1 video boundary similarity at 0.92–0.97 (I2V) across ten chunks of matched ≈ 53 -second single-scene clips, held the high-frequency energy share flat end-to-end in both conditioning modes, and held audio loudness to its chunk-1 reference against a raw +72% drift (T2V). A two-stage pipeline (streaming Stage 1 + windowed distilled Stage 2 with per-window detail anchoring and frozen audio) delivers a fully refined 2 $\times$ -resolution audio-video output in ≈ 98 minutes per clip on a single 16 GB GPU, essentially independent of conditioning mode.

Running matched T2V and I2V generations under an identical seed and guidance stack, differing only in the presence of a guide image, shows that visual conditioning is not a lever on streaming audio behaviour: chunk-1 audio coherence, envelope repetition, and loudness stability all track the prompted sound content rather than the guide image, while the guide image’s clear effects are on the video side (a stable anchored scene with a 0.92 boundary floor versus T2V’s early-run 0.38–0.73 hand-offs under sustained motion). Two failure modes persist across both runs at this horizon and are characterised rather than resolved: a per-chunk audio loudness gesture shown by live intervention to be structural to chunked autoregressive generation over a static prompt — immune to continuity buffers, guidance weights, and every tested

perturbation of the audio reference, and removable only by a non-chunked audio pass that current VRAM budgets do not admit — and a chunk-length-invariant ≈ 4 -second video layout oscillation that appears intrinsic to the model’s trained temporal window. All of this is visible in the pipeline’s per-chunk telemetry — unconditional settings dumps, drift-event tables, and a phase-lock detector — which is designed to localise such effects in wall-clock time and latent-frame position, and to make a silently inactive guidance path impossible to miss (a guidance term that does not change per-step wall-clock time is not running).

Object-level identity persistence across chunk boundaries, invisible to the current structural metrics in both conditioning modes, remains the principal open problem for content-level quality.

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